

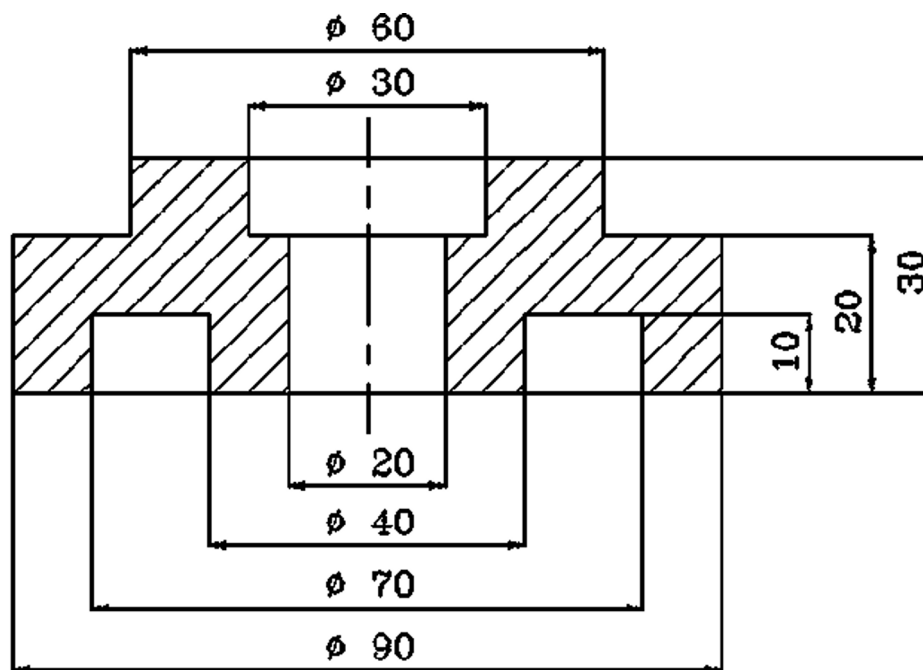


Example 1



The figure shows a component to be obtained through a sand casting and machining processes. The component will be made of grey cast iron. The pattern will be made of wood.

After having appropriately chosen the parting line, design the pattern and any cores respecting machining allowance (3 mm), draft angles (4°), fillets/radii (3 mm), and shrinkage allowance (1%). Suppose that all surfaces need to undergo subsequent machining.



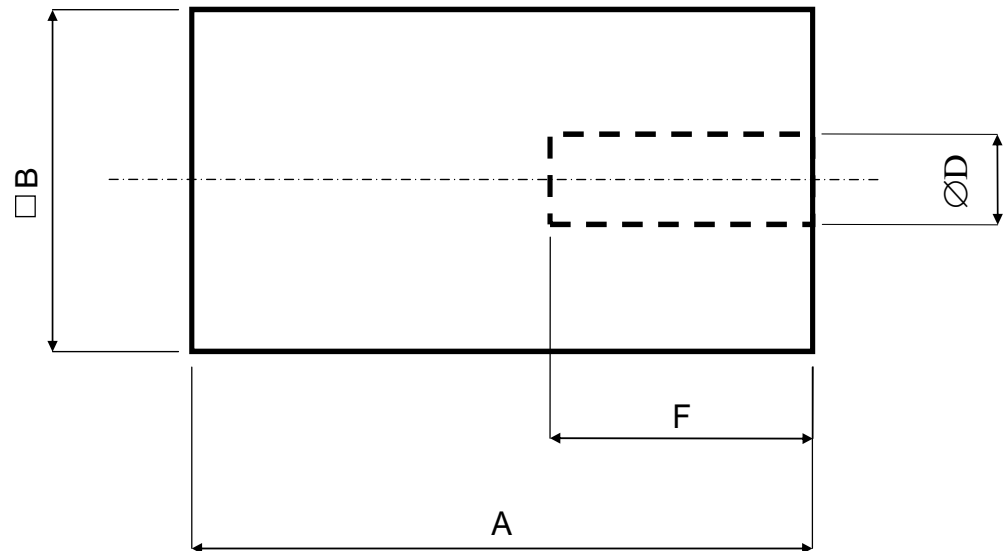
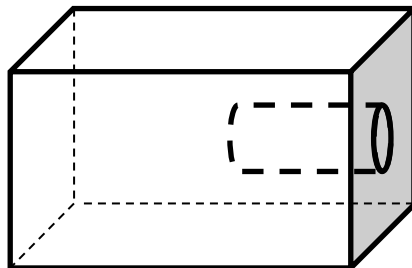


Example 2

The component in the figure must be made by sand casting.

Dimensional data: $A = 300$ mm, $B = 100$ mm, $D = 30$ mm, $F = 150$ mm.

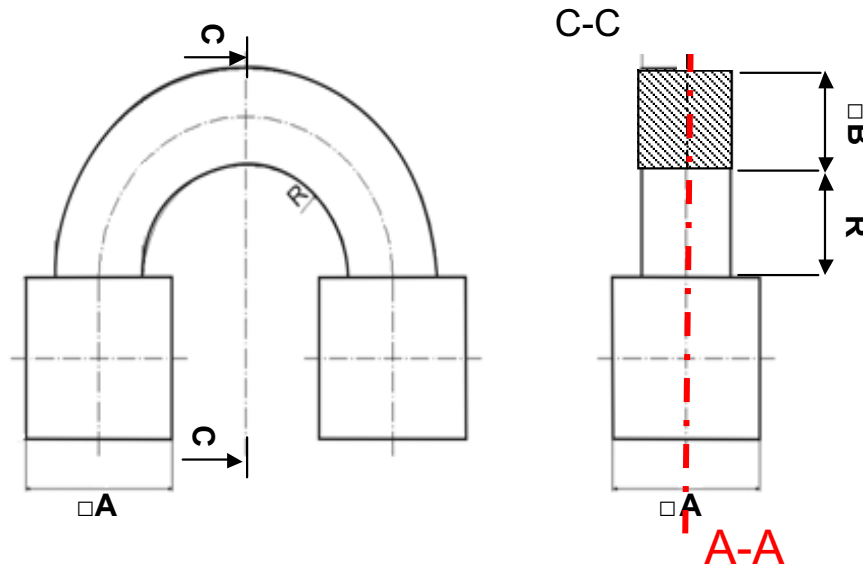
- Knowing that the hole will have to undergo machining, design the pattern assuming a material allowance of 2 mm where necessary, and a shrinkage allowance of 1%
- If you wish to use a cylindrical core print with a diameter of 70 mm, determine its minimum length to ensure its stability during the forming process





Example 3

The pattern shown will be used to create the cavity in sand casting.
The plane A-A is chosen as the parting line of the mould.
Please, study the directional solidification of the metal.
Dimensional data: $A = 31 \text{ mm}$, $B = 20 \text{ mm}$, $R = 20 \text{ mm}$.

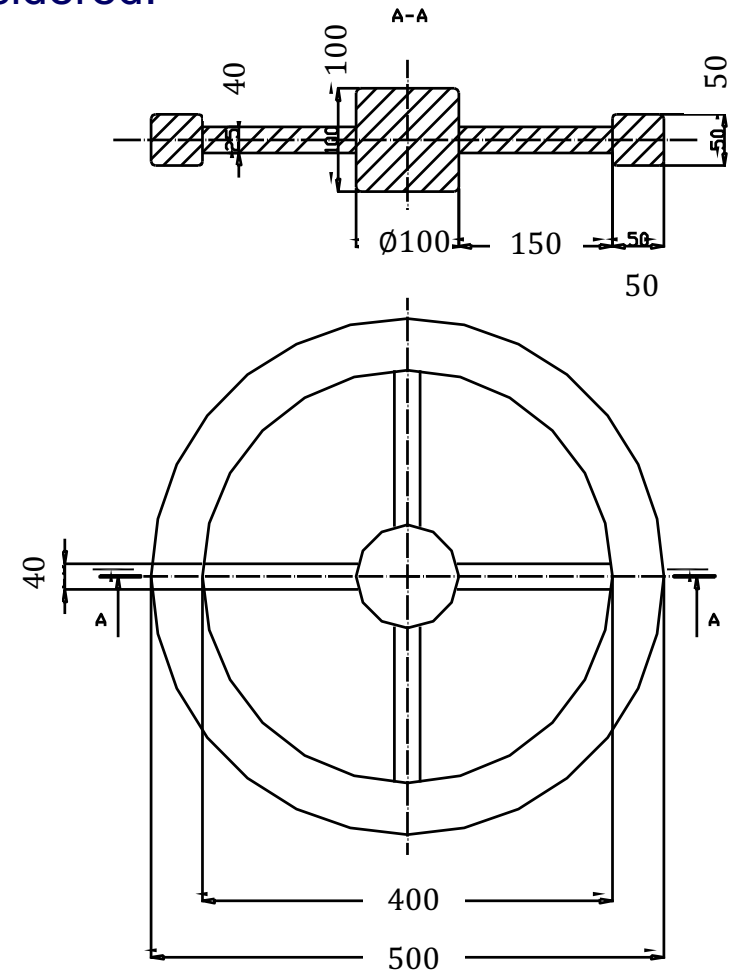
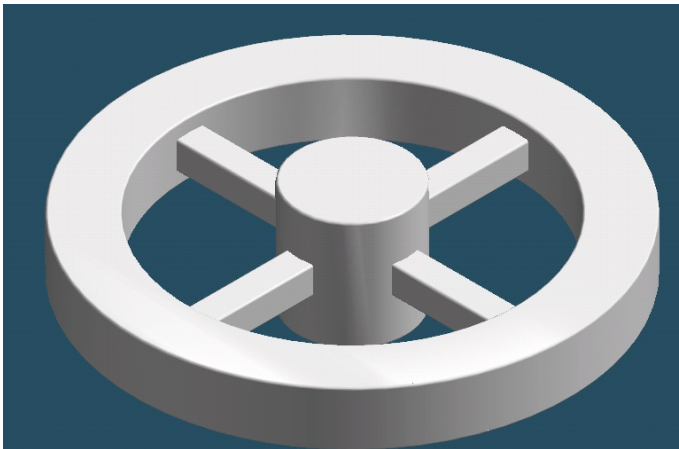




Example 4

Study the directional solidification of the casting shown in the figure.

Knowing that a 68 mm diameter hole will be drilled in the hub, if needed an up to 64 mm diameter cylindrical core may be considered.

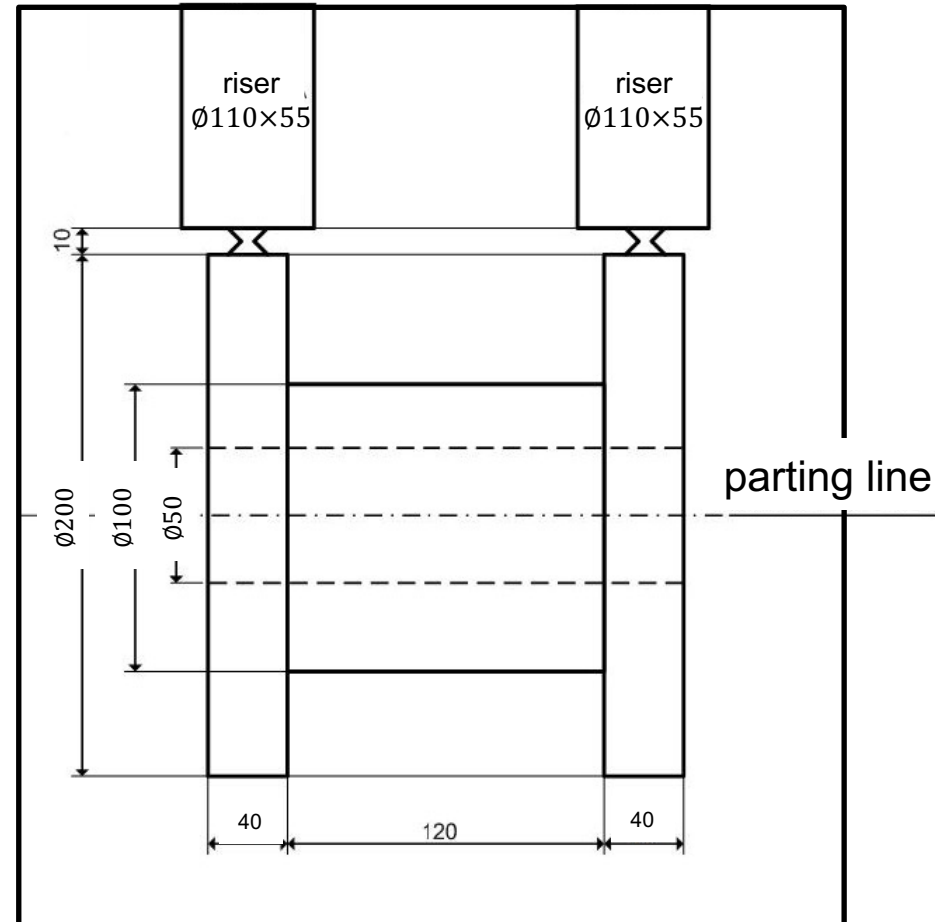




Example 5



The figure shows a mould cavity with risers used in a sand casting process. A middle gating system is considered with two gates, one runner and one sprue (middle gating system = gating system with gates on the parting line). Design the gating system to ensure a mould filling time equal to 6 s. Consider $A_s:A_r:A_g = 2:2:1$ and $c = 0,55$.





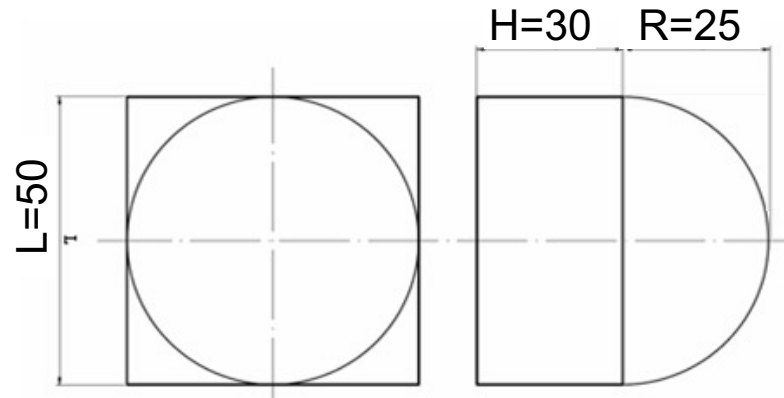
Example 6



The pattern shown will be used to create a cavity in sand casting.

A bottom gating system has to be design to ensure a mould filling time equal to 5 s and $A_s:A_r:A_g = 3:3:2$. The height of the cope is equal to 75 mm.

- Design the gating system considering $c = 0,5$ and a blind riser on the parting line with a volume equal to 50 cm^3 .
- If a gate has a cross-section equal to $0,25 \text{ cm}^2$, complete the detailed design of the gating system and determine the actual mould filling time.





Example 7



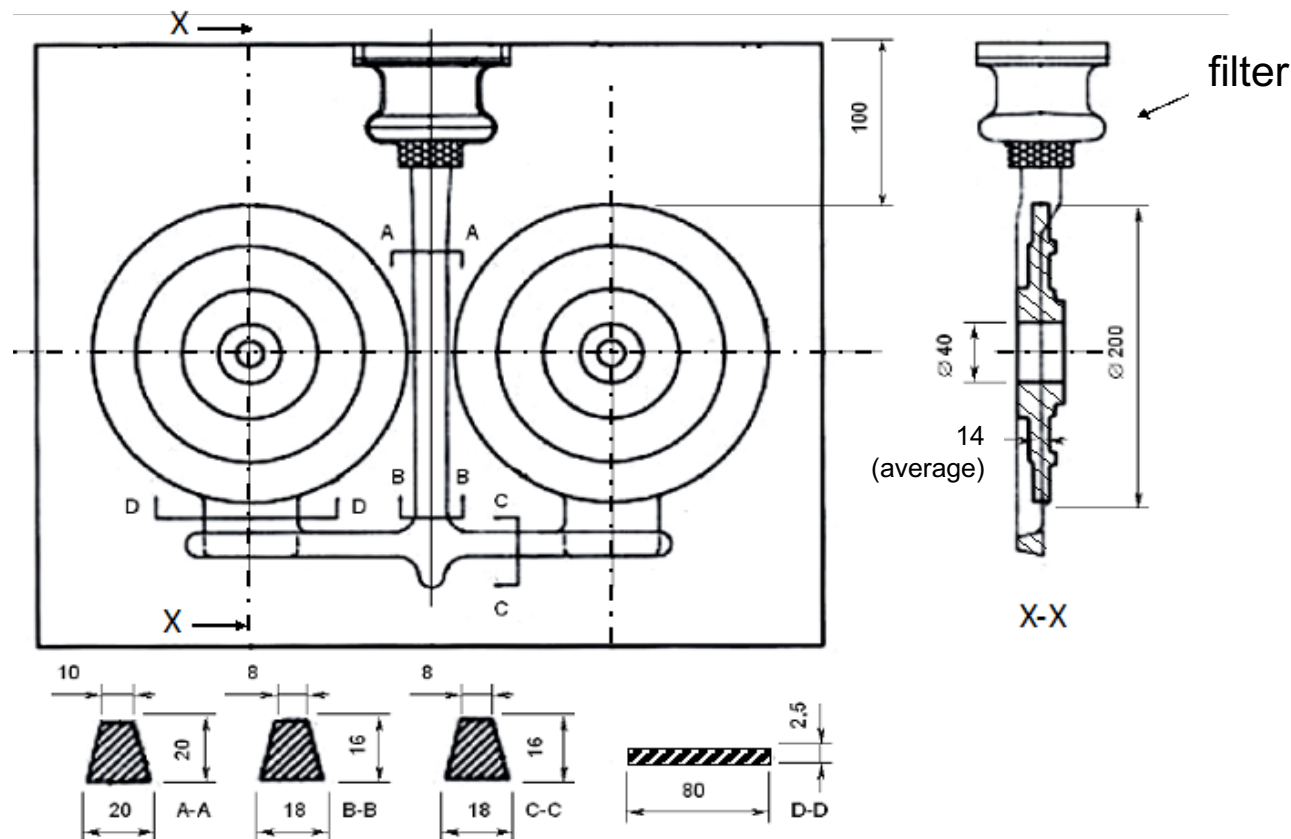
A foundry is casting brake discs for the automotive sector.

The figure shows the pattern used to shape a mould with vertical parting line.

Consider an average thickness of the disc equal to 14 mm, two blind risers with a volume equal to $1,5 \cdot 10^5 \text{ mm}^3$ each and a height smaller than 100 mm.

Determine the V/A ratio for the discs.

Considering a disc volume equal to $5 \cdot 10^5 \text{ mm}^3$, compute the actual mould filling time ($c = 40\%$) and compare it with a time constraint equal to 10 s.



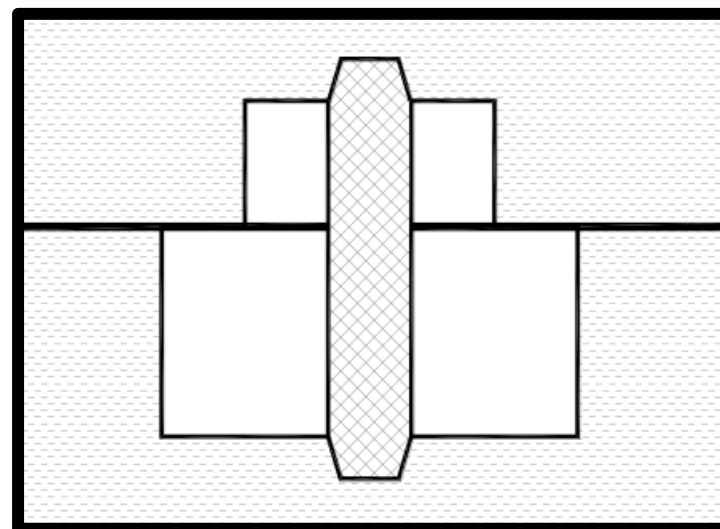
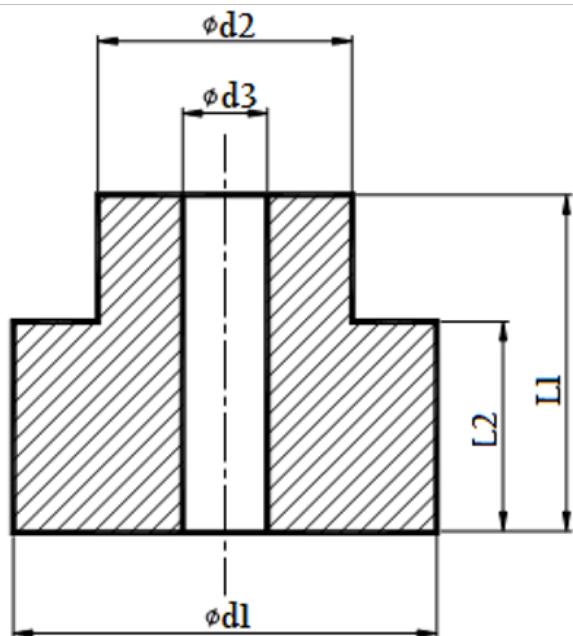


Example 8



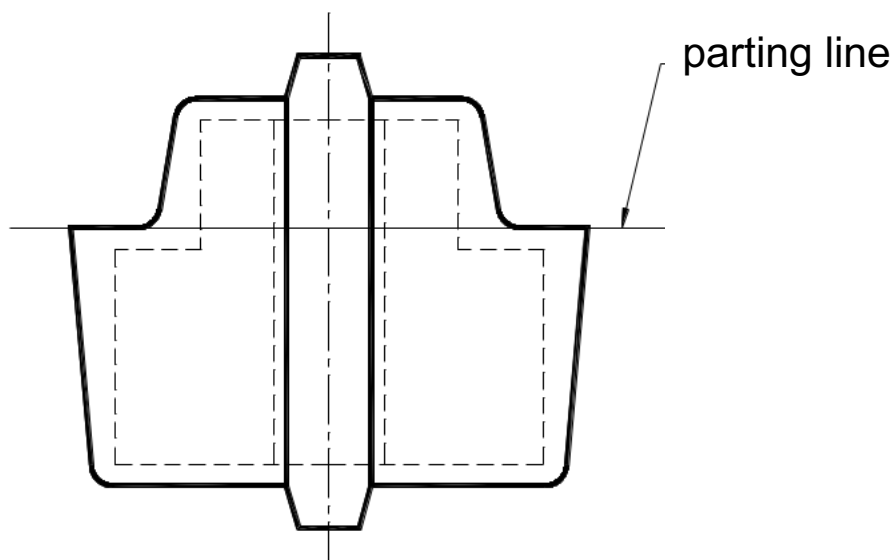
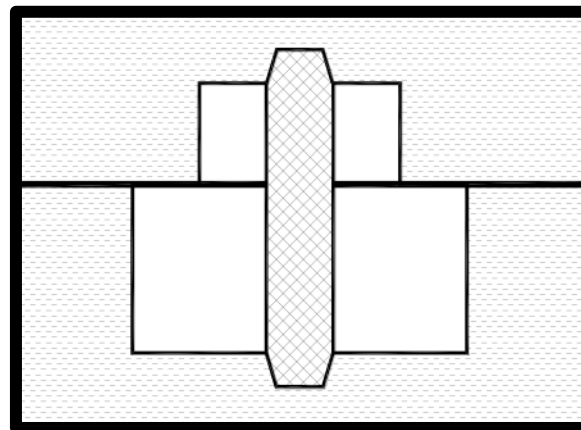
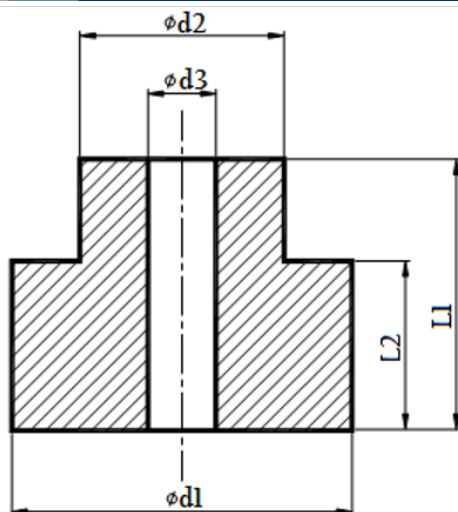
The figure shows a component to be obtained through a sand casting and machining processes. The component will be made of grey cast iron. The pattern will be made of wood. Dimensional data: $d1 = 100$ mm, $d2 = 50$ mm, $d3 = 20$ mm, $L1 = 120$ mm, $L2 = 70$ mm.

Design the pattern and the core respecting machining allowance (4 mm), draft angles (4°), fillets/radii (3 mm), and shrinkage allowance (1%). Suppose that all surfaces need to undergo subsequent machining.





Example 8 - Solution





Example 9



The figures show a component to be obtained through a sand casting. The plane A-A is chosen as the parting line of the mould. Design a middle gating system to ensure a mould filling time not greater than 10 s (*do not consider risers and shrinkage allowance*).
Data: $L = 2 \text{ cm}$, $H_{\text{cope}} = 2L$, $H_{\text{drag}} = 3L$, $A_s:A_r:A_g = 4:2:1$, $c = 0,5$.

